Implementation of the Dehbi (2008) Continuous Random Walk Model as an ANSYS Fluent UDF

Technical report: aerosol deposition in a 90° pipe bend

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Abstract

The continuous random walk (CRW) model of Dehbi (2008), which integrates a normalised Langevin equation for the fluid velocity seen by a particle, was implemented as a compiled ANSYS Fluent user-defined function (UDF). It was applied to aerosol deposition in a 90° pipe bend (diameter 20 mm, curvature ratio 4, Reynolds number 4.1 × 10³) on a Reynolds-stress-model flow field, with 10⁴ particles per size class. The computed deposition fraction increases from 16.7 % for the smallest class (nominally 10 µm; Stokes number 0.09) to 98.0 % at 50 µm (Stokes number 2.3). At equal Stokes number it differs from the bend correlation of Pui et al. (1987) by −5.5 to +1.4 percentage points, but at the smallest Stokes number it exceeds the published bend measurements; because the geometry and flow conditions differ from those of the experiments, this comparison is qualitative. Source inspection and a statistical test identify five departures from the published model: the near-wall statistics and the drift correction act only in wall-adjacent cells, the bulk drift is omitted, the Langevin state is not transformed between reference frames, successive random deviates are correlated, and the one-radius deposition criterion is not applied.

1. Introduction

Discrete random walk (DRW) models sample velocity fluctuations from locally isotropic turbulence. In inhomogeneous near-wall turbulence they produce a spurious drift of low-inertia particles towards the wall, which violates the well-mixed condition (Thomson, 1987) and leads to over-prediction of small-particle deposition (Dehbi, 2008). Dehbi (2008) replaced the DRW with a normalised Langevin equation that uses anisotropic near-wall rms profiles and a drift term proportional to the wall-normal gradient of the rms velocity. This report documents an implementation of that model as a compiled ANSYS Fluent UDF, its application to a 90° pipe bend for particle diameters of 10 to 50 µm, a comparison with published bend data, and the points at which the implementation departs from the published model.

2. Case set-up

Table 1. Geometry, flow conditions and solver settings.

Item	Value
Pipe diameter D; curvature ratio $R_0 = R/r$ (bend radius over pipe radius)	20 mm; 4 (R = 40 mm)
Straight inlet and outlet sections	$5D$ each
Fluid	air, ρ = 1.225 kg m⁻³, μ = 1.81 × 10⁻⁵ Pa s
Inlet; outlet	uniform velocity U = 3.0 m s⁻¹; pressure outlet
Reynolds number $\rho U D/\mu$; friction Reynolds number $u_\tau r/\nu$	4.06 × 10³; ≈ 140 (Blasius estimate)
Mesh	structured hexahedral, ≈ 5.3 × 10⁶ cells, wall $y^+ \approx 1$
Solver	ANSYS Fluent 2024 R2, pressure based, steady, SIMPLE
Turbulence model	Reynolds stress model (RSM), linear pressure–strain, enhanced wall treatment

Item	Value
Discretisation	least-squares gradients, second-order pressure, second-order upwind
Discrete phase	inert spheres, one-way coupling, steady tracking, 10^4 particles per size class, built-in dispersion model disabled
Particles	density 1000 kg m^{-3}, released at the gas velocity (UDF header; see Section 6)
Deposition	particle stopped at first wall contact (reported by Fluent as “aborted”)



Figure 1. Computational domain: straight inlet of length 5D, 90° bend of curvature ratio R_0 = 4 and straight outlet of length 5D; D = 20 mm.

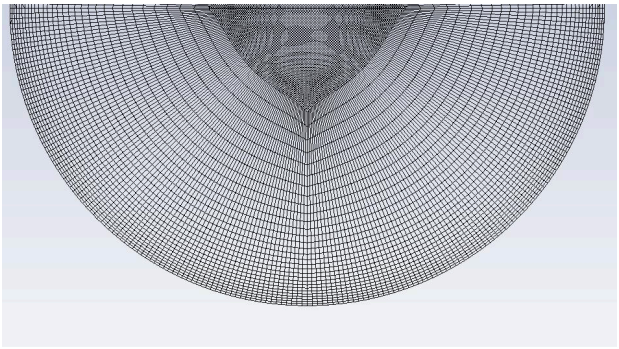
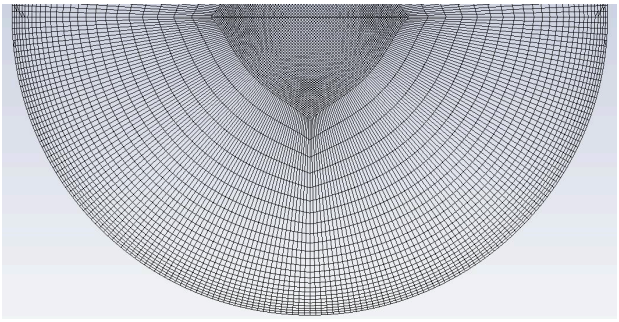
Refined mesh, 5.3 × 10^6 cells	Baseline mesh, 2.5 × 10^6 cells
	

Figure 2. Bend cross-sections of the refined and baseline meshes (screenshots not at the same scale). The refined mesh was used for all results; results on the baseline mesh are not archived.

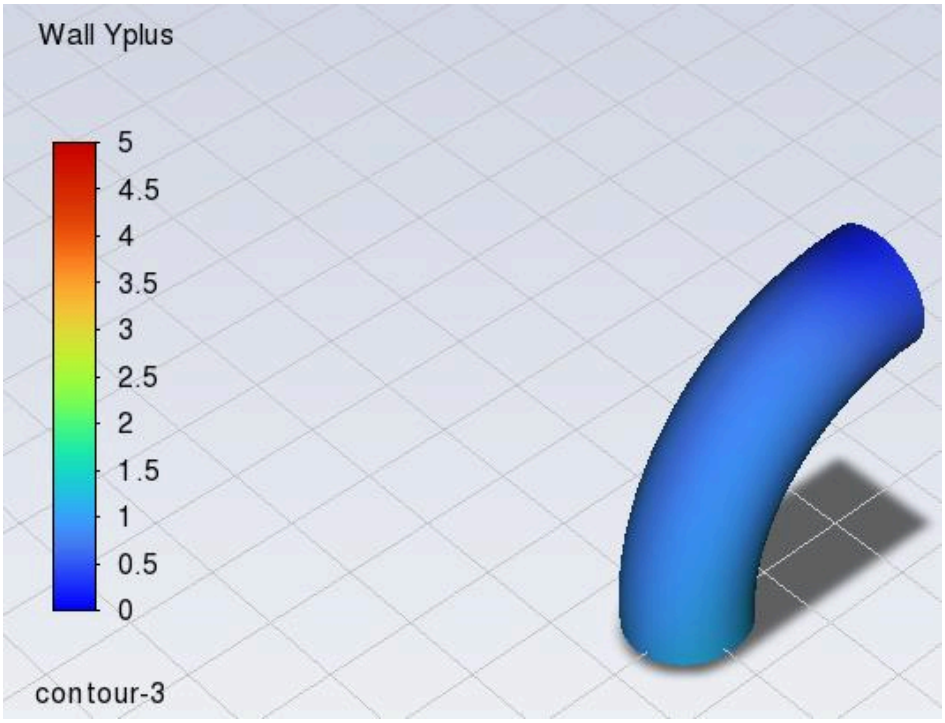


Figure 3. Wall y^+ on the bend surface; the values lie below about 1.2 (colour scale 0–5).

After about 950 iterations the scaled residuals of the continuous phase were between 7×10^{-4} (ε) and 2×10^{-6} and still decreasing slowly (`figures/residuals.png`; the legend of the archived screenshot is truncated). No integral-quantity monitors were archived. The Fluent dialogs for the turbulence model, solution methods, discrete-phase numerics, UDF hooks and injection, the start of Fluent from the Visual Studio command prompt used to compile the UDF, and the DPM console summaries of the five runs are archived in `figures/setup/`.

3. Model and implementation

Line numbers refer to `udf/crw_dehbi2008_rev10.c`. Its executable code is identical to the file first uploaded to this repository (commit `dfa9cad`, May 2026); only comments have been revised since.

Coupling. Fluent integrates the particle momentum equation with the mean slip velocity. The UDF adds the drag of the fluid velocity fluctuation u_i' as a body force per unit mass (lines 644–738):

$$\frac{F_i}{m_p} = f_D \, u_i', \qquad f_D = \frac{1 + 0.15 \, Re_p^{0.687}}{\tau_p}, \qquad \tau_p = \frac{\rho_p d_p^2}{18 \, \mu},$$

where Re_p is the particle Reynolds number based on the mean slip velocity. If Fluent's default spherical drag law (Morsi and Alexander, 1972) was used for the mean drag, the two drag factors differ by up to about 6 % for $0.1 < Re_p < 1$; the drag law selected in the case is not recorded.

Langevin equation. The normalised fluctuation $\eta_i = u_i'/\sigma_i$, where σ_i is the rms fluid velocity, obeys $d\eta_i = -\eta_i \, dt/\tau_L + A_i \, dt + \sqrt{2/\tau_L} \, dW_i$. Here τ_L is the Lagrangian integral time scale of the fluid seen by the particle, A_i is a drift term and dW_i is a Wiener increment. The equation is advanced with a semi-implicit Euler–Maruyama scheme (lines 397–399, 427–429), in which ξ_i are standard normal deviates (Section 5):

$$\eta_i^{n+1} = \frac{\eta_i^n + \sqrt{2 \, \Delta t/\tau_L} \, \xi_i + A_i \, \Delta t}{1 + \Delta t/\tau_L}.$$

Near-wall branch (wall-adjacent cells). Dehbi (2008) uses anisotropic statistics for $y^+ < 100$ in a body-fitted frame. In the UDF, e_2 is the wall normal pointing into the fluid, e_1 the local mean-flow direction made orthogonal

to e_2 (streamwise) and $e_3 = e_1 \times e_2$ (spanwise) (lines 225–275). The on-demand function `init_wall_udm` (lines 498–625) stores the friction velocity u_τ, the y^+ of the cell centroid and the wall normal for cells adjacent to a wall only. Every other cell is assigned $y^+ = 101$ at run time and uses the bulk branch (lines 375–383). The near-wall branch is therefore active only in the first cell layer, where $y^+ \approx 0.5$–1 on the present mesh. Its rms profiles are the DNS-based fits given by Dehbi (2008) (lines 131–148):

$$\frac{\sigma_1}{u_\tau} = \frac{0.40\, y^+}{1 + 0.0239\, (y^+)^{1.496}},$$

$$\frac{\sigma_2}{u_\tau} = \frac{0.0116\, (y^+)^2}{1 + 0.203\, y^+ + 0.00140\, (y^+)^{2.421}},$$

$$\frac{\sigma_3}{u_\tau} = \frac{0.19\, y^+}{1 + 0.0361\, (y^+)^{1.322}}.$$

The Lagrangian time scale follows Kallio and Reeks (1989): $\tau_L^+ = \tau_L u_\tau^2/\nu = 10$ for $y^+ \le 5$ and $\tau_L^+ = 7.122 + 0.5731\, y^+ - 0.00129\, (y^+)^2$ otherwise (lines 196–214). Only the wall-normal component carries a drift, reduced by the local Stokes number $St_L = \tau_p/\tau_L$; its derivative is evaluated analytically (lines 163–188, 421–422):

$$A_2 = \frac{1}{1 + St_L}\, \frac{\partial \sigma_2}{\partial y}, \qquad A_1 = A_3 = 0.$$

Bulk branch (all other cells). Isotropic turbulence with $\sigma = (2k/3)^{1/2}$ and $\tau_L = (2/C_0)\, k/\varepsilon$, $C_0 = 14$, where k is the turbulent kinetic energy and ε its dissipation rate (lines 444–474). Dehbi (2008) includes a drift proportional to ∇k; it is omitted here because accessing the cell gradient of k under the RSM caused a solver failure. Between the second cell layer ($y^+ \approx 1.5$) and $y^+ = 100$, about 70 % of the pipe radius, the wall-normal fluctuation is therefore modelled with the isotropic value and without drift. With k taken from the fits above, the isotropic value exceeds σ_2 by factors of about 7 at $y^+ = 5$ and 2 at $y^+ = 30$. The well-mixed condition is not enforced in this region, and because the extent of the near-wall branch is set by the first-cell height, the model itself depends on the near-wall mesh.

Wall interaction and time step. `dehbi_wall_bc` (lines 749–752) stops a particle when its trajectory reaches a wall face. The particle time step is limited to $\min(10^{-6}\ \mathrm{s},\ 0.1\ \tau_p)$ (lines 765–797), which is 10^{-6} s for all diameters considered.

Table 2. Departures of the implementation from the published model.

Element	Dehbi (2008)	Present implementation
Near-wall statistics and drift	applied for $y^+ < 100$	applied only in wall-adjacent cells, at the cell centroid
Bulk drift	proportional to ∇k	omitted
Frame of the Langevin state	body-fitted near the wall, Cartesian in the bulk	the same, but the state is not transformed when a particle changes branch
Random forcing	independent Gaussian increments	deviates hashed from particle address and time; serially correlated (Section 5)
Deposition	particle centre within one particle radius of the wall	particle trajectory reaches a wall face

4. Results

The Stokes number is $Stk = 2\tau_p U/D = \rho_p d_p^2 U/(9\mu D)$, based on the bulk velocity and the pipe radius as in Pui et al. (1987), and is evaluated with ρ_p = 1000 kg m^{-3} and without Cunningham correction. The deposition fraction is DF = N_{dep}/N, where N is the number of tracked particles and N_{dep} the number reported as "aborted"; it equals the deposition efficiency (one minus the penetration) of Pui et al. (1987) and may include deposition in the straight sections (Section 6). The binomial standard error $\sqrt{DF(1-DF)/N}$ assumes statistically independent particle histories; because the generator correlates particles (Section 5), it is a lower bound.

Table 3. Deposition fraction (DF) from the Fluent DPM summaries (`figures/setup/dpm-summaries.png`; data in `data/deposition_fraction.csv`), and the correlation of Pui et al. (1987) at the same Stokes number.

d_p (µm)	Stk	Tracked	Escaped	Deposited	DF (%)	Standard error (%)	Pui et al. (%)
10[a]	0.092	10,000	8,329	1,671	16.7	0.37	18.5
20	0.368	10,000	4,279	5,721	57.2	0.49	55.8
30	0.829	10,000	2,142	7,858	78.6	0.41	84.1
40	1.473	10,000	546	9,454	94.5	0.23	96.2[b]
50	2.302	10,000	199	9,801	98.0	0.14	99.4[b]

[a] Nominal diameter; see Section 6. [b] Beyond the measured range of the correlation.

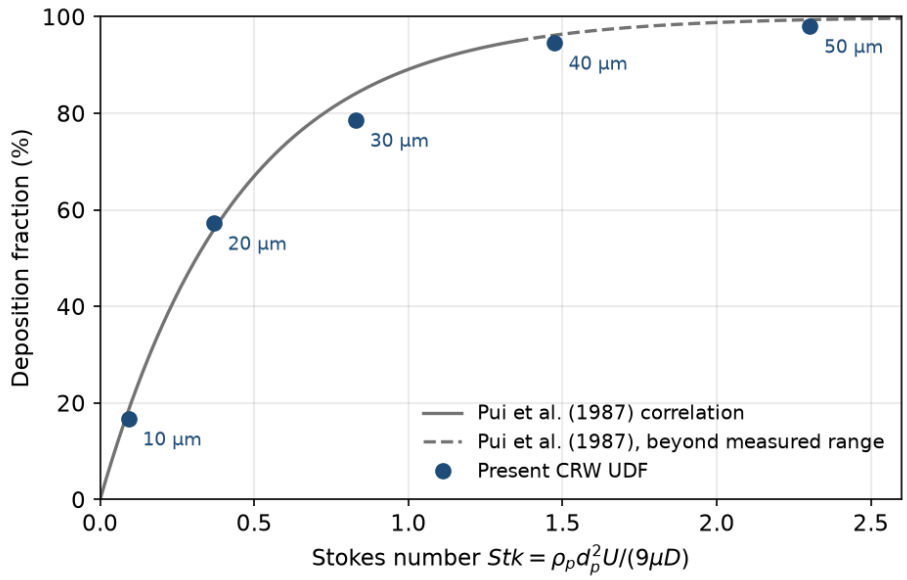


Figure 4. Deposition fraction against Stokes number (ρ_p = 1000 kg m^{-3} assumed), with the correlation of Pui et al. (1987) for R_0 = 5.7 and Re = 6000–10,000, drawn dashed beyond the measured range. Binomial 95 % intervals are smaller than the markers; the 10 µm label is nominal (Section 6). Generated by `tools/plot_deposition.py`.

The deposition fraction increases monotonically with Stk, from 16.7 % at Stk = 0.09, where particle inertia is small, to near-complete deposition at $Stk > 1.5$. The correlation of Pui et al. (1987), $DF = 1 - 10^{-0.963\,Stk}$, was fitted to measurements in 90° bends with tube diameters of 5.03–8.51 mm, R_0 = 5.7 and Re = 6000–10,000; the largest measured Stokes number is about 1.35 (Sippola and Nazaroff, 2002, Eq. 121, Table 8 and Fig. 6). At equal Stk the computed deposition fractions differ from the correlation by −5.5 to +1.4 percentage points. The correlation does not follow every measurement, however. The same figure shows the measurements of McFarland et al. (1997) in 90° bends of 16 mm diameter with R_0 = 1–20 and Re = 3000–20,000, which include the present curvature ratio. Read from that figure (approximate values), the deposition fraction measured in both studies is

about 7–10 % at Stk = 0.09, below the present 16.7 %, and about 55–60 % at Stk = 0.37, close to the present 57.2 %; for R_0 = 4 it reaches about 90 % at $Stk \approx 0.67$, above the present 78.6 % at Stk = 0.83. The present case differs from the experiments of Pui et al. in curvature ratio (4) and Reynolds number (4.1 × 10³), and its inlet condition (uniform velocity $5D$ upstream of the bend) gives a flow that is still developing at the bend; its deposition fraction may include the straight sections, and its two largest Stokes numbers lie beyond the range of the measurements. With ρ_p = 1550 kg m⁻³, the default density of the material named in the archived injection dialog, the correlation would give 27.1, 71.8 and 94.2 % at the three smallest diameters. Given these differences and the departures in Table 2, the comparison is qualitative and is not a validation of the model.

5. Random-number generator test

The function `gauss_rand` (lines 94–121) generates Box–Muller deviates from a hash of the particle memory address and the particle time, without a stored state. The program `tools/rng_check.c` runs a functionally identical copy (types adapted to run outside Fluent) at the time step of the present runs.

Table 4. Statistics of the UDF generator at Δt = 10⁻⁶ s (output of `tools/rng_check.c`; the test number is given in brackets; test [3], for six fixed addresses, is summarised by test [4]).

Statistic	Ideal	Result
[1] Mean, variance (2 × 10⁵ samples)	0, 1	+0.001, 1.003
[1] Autocorrelation, lags 1 to 5	0	+0.22, +0.06, −0.11, −0.29, −0.23
[2] Correlation of two particles at equal time, address gaps 8 B, 32 B, 448 B, 8 KiB	0	+0.85, −0.49, +0.13, −0.65
[4] Normalised variance of the sum of 1000 successive deviates, 2000 particle addresses	1	median 0.48; 5–95 % range 0.20–2.7; maximum 60
[5] Time average of η^2 in the Langevin update over 0.044 s (τ_L = 3.32 ms, τ_L^+ = 10), 2000 particle addresses	1 (ideal generator: median 0.94, 5–95 % range 0.51–1.70)	median 0.41; 5–95 % range 0.15–2.9; maximum 465

The mean and variance agree with those of a standard normal distribution. Because the particle time advances in equal steps, successive deviates of one particle are correlated, and deviates of two particles at the same time can be strongly correlated. Driven by this generator, the Langevin update gives a time-averaged variance of the modelled fluid velocity below its target for most particles and far above it for a few. The random forcing, and with it the effective turbulent diffusivity, therefore varies between particles, and the standard errors in Table 3 underestimate the statistical uncertainty.

6. Limitations and further work

Recorded settings. The archived injection dialog for the smallest size class (`figures/setup/injection.png`) specifies d_p = 5 µm, the material “anthracite” and zero initial velocity, whereas Table 3 assumes 10 µm and 1000 kg m⁻³ and the UDF header specifies release at the gas velocity; at 5 µm, Stk for this class would be 0.023 (0.036 at 1550 kg m⁻³). Gravity, the inlet turbulence, the drag law, the DPM step limit, the solver precision and the assignment of `dehbi_wall_bc` to the three wall zones (inlet pipe, bend, outlet pipe) are not recorded in the archived dialogs, and deposition was not recorded per wall zone.

Implementation. The departures in Table 2 are expected to matter most for the smallest particles, whose deposition depends on turbulent dispersion rather than on inertial impaction. Whether Fluent reuses particle memory between particles, which would make particles share random sequences, has not been tested.

Further work. A quantitative assessment of the model requires (i) a per-particle random-number state, (ii) near-wall statistics evaluated at the particle position throughout $y^+ < 100$, (iii) a consistent transformation of the Langevin state between frames, (iv) a well-mixed tracer test, (v) the confirmed injection settings, and (vi) a quantitative comparison with bend measurements at matched curvature ratio and Reynolds number, such as those of McFarland et al. (1997) at R_0 = 4.

7. Conclusions

The CRW model of Dehbi (2008) was implemented as a compiled ANSYS Fluent UDF and applied to a 90° pipe bend at Re = 4.1 × 10³. The computed deposition fraction increases from 16.7 % at Stk = 0.09 to 98.0 % at Stk = 2.3 and differs from the correlation of Pui et al. (1987) by −5.5 to +1.4 percentage points at equal Stokes number, while exceeding the measured bend deposition at the smallest Stokes number. The implementation departs from the published model in the five respects listed in Table 2, most importantly in that the near-wall statistics and drift are applied only in wall-adjacent cells and the bulk drift is omitted. The results show that the model can be coupled to Fluent's discrete phase model through body-force and boundary-condition UDFs; a quantitative test of the model requires the changes listed in Section 6.

Software

The Cursor code editor was used during UDF development to debug C syntax, memory-allocation and compilation errors related to the Fluent UDF macros, and to tidy and comment the code before compilation; the physical model, its equations and the coupling follow Dehbi (2008). The PDF version of this report is generated from the Markdown source with pandoc and Chromium.

References

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